Arithmetic Classification and Multiplicative Monoid Structure of Natural Numbers: A Seven-Class Partition, Prime Coordinate Representation, and Deterministic Addition Laws

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Version 1.2 — April 16, 2026

Archived on Zenodo (Concept DOI, always resolves to latest version):
doi.org/10.5281/zenodo.19511136

Abstract

The natural numbers are conventionally treated as ordered magnitudes. This paper develops a complementary framework: every natural number n encodes intrinsic structural information through its prime factorization—information that is independent of the quantity n represents and computable in deterministic, finite time.

We make five contributions, three of which systematize classical results into a unified framework and two of which are, to our knowledge, novel.

Systematization. **(C1)** We define a seven-class arithmetic partition of $\mathbb{N}$ based on two invariants—the total prime factor count $\Omega(n)$ and the residue class of primes modulo 6—and prove it forms a finite commutative monoid $(S, \otimes)$ under induced multiplication, with a complete 49-entry Cayley table. **(C2)** We characterize the power dynamics $\mathrm{cl}(n^k)$ and prove the Universality Theorem: every $n > 1$ reaches the absorbing class S_6 in at most 3 steps. **(C5)** We define the resonance relation $R(i, j) = \Omega(\gcd(i, j))$ and prove that $\approx 61\%$ of integer pairs are non-resonant (coprime), matching the classical value $6/\pi^2$.

Novel contributions. **(C3)** We identify exactly 7 of the 28 unordered class pairs that yield a deterministic addition law: six are trivially forced by the singleton classes ($S_0 = \{1\}$, $S_3 = \{2\}$, $S_4 = \{3\}$), and one— Solar + Lunar $\to S_6$—is the unique *non-trivially* deterministic law (Theorem 4.4, verified on 90,000 pairs). **(C4)** We formalize the SpectralAddress—a K-dimensional vector of p-adic valuations, lossless for $\mathbf{B}$-smooth integers—and prove four $O(K)$ isomorphisms mapping multiplication, GCD, LCM, and coprimality testing to elementary vector operations.

All results are computationally verified in Python (100,000 pairs per foundational axiom), C++20 ($\geq$475 M Cayley lookups/s on Intel Core i5; 559 M verified on AMD Ryzen 9 3900), and a Verilog RTL module intended for synthesis, validated via Python behavioral emulation (14/14 test vectors; on-device synthesis and timing simulation pending).

Keywords: prime factorization, arithmetic classification, multiplicative monoid, p-adic valuation, SpectralAddress, resonance, deterministic addition laws, FPGA number theory

Repository: github.com/Prime-Spectrum-Team/Numbers-are-Information-

1 Introduction

1.1 The Standard View and Its Limitation

In computing, integers serve as universal indices: array positions, memory addresses, and sequence positions are identified by integers whose sole operationally relevant property is their ordinal position.

This view is not wrong, but it is incomplete.

The prime factorization of any integer $n > 1$ encodes structural information that is:

- **Intrinsic**—present in the number itself;
- **Deterministic**—computable by a fixed procedure;
- **Compositional**—the structure of a product is determined by the structures of the factors.

Consider $n = 60 = 2^2 \times 3 \times 5$. This factorization assigns it depth $\Omega(60) = 4$ and places it at the intersection of three prime axes. These are not properties *about* 60; they are properties 60 *is*.

1.2 Complementary Viewpoint

The standard view treats n as an ordinal. The complementary view developed here treats n as a coordinate in prime space: 60 has SpectralAddress $[2, 1, 1, 0, 0, 0, 0]$, placing it at the intersection of prime axes 2, 3, and 5. Both views are consistent; they are adapted to different purposes. This prime-arithmetic view has a concrete operational consequence: every arithmetic operation on **B**-smooth integers reduces to a corresponding linear operation on SpectralAddress vectors—all in $O(K)$ time.

1.3 Contributions

C1 Seven-class partition and multiplicative monoid (Theorems 3.5–3.7). *Systematization.*
C2 Power dynamics and universality (Theorem 3.10). *Systematization.*
C3 Exactly 7 deterministic addition laws; 1 non-trivially deterministic (Theorem 4.4). *Novel.*
C4 SpectralAddress and four $O(K)$ isomorphisms (Propositions 5.2–5.4). *Novel.*
C5 Resonance and $6/\pi^2$ sparsity (Theorem 6.2). *Systematization.*

1.4 Related Work

The individual components—Ω-function [5, 8], prime factor count distributions [6, 7], p-adic valuations [4]—are classical. A detailed positioning relative to prior work is given in Section 9.

2 Preliminaries

2.1 The Ω-Function

Definition 2.1. *For $n \in \mathbb{N}$ with canonical factorization $n = \prod_i p_i^{a_i}$, define $\Omega(n) = \sum_i a_i$ for $n > 1$ and $\Omega(1) = 0$.*

Proposition 2.2 (Complete additivity of Ω)**.** *For all $a, b \in \mathbb{N}$: $\Omega(ab) = \Omega(a) + \Omega(b)$.*

Proof. From unique prime factorization: $\Omega(ab) = \sum(\alpha_i + \beta_i) = \Omega(a) + \Omega(b)$. □ □

This single axiom generates all 49 multiplication laws of the monoid. The Erdős–Kac theorem [2] establishes that $\Omega(n)$ is asymptotically normally distributed with mean and variance $\ln \ln n$, confirming that S_6 dominates at large N.

Verification (E1.1). 100,000 random pairs from $[1, 10^6]$: success rate 1.0 (95% CI: [0.99996, 1.0]).

2.2 Prime Residues Modulo 6

Proposition 2.3. *Every prime $p > 3$ satisfies $p \equiv 1 \pmod 6$ or $p \equiv 5 \pmod 6$.*

Proof. Among $\{0, 1, 2, 3, 4, 5\}$ mod 6: residues $\{0, 2, 4\}$ imply divisibility by 2; residue 3 implies divisibility by 3. For prime $p > 3$, only residues 1 and 5 remain. □ □

By Dirichlet's theorem [1], these two residue classes contain asymptotically equal numbers of primes. *Verified on all 664,579 primes below 10^7: ratio $= 0.992$.*

2.3 The p-adic Valuation

Definition 2.4. *For prime p and $n \in \mathbb{N}$, the p-adic valuation $\nu_p(n) = \max\{a \geq 0 : p^a \mid n\}$.*

Key property: $\nu_p(ab) = \nu_p(a) + \nu_p(b)$.

3 The Seven-Class Partition and Multiplicative Monoid

3.1 Definition and Interpretation

Definition 3.1 (PrimeSpec classification)**.** *Define* cl : $\mathbb{N} \to S = \{S_0, S_1, S_2, S_3, S_4, S_5, S_6\}$*:*

$$\mathrm{cl}(n) = \begin{cases} S_0 & n = 1 \text{ (Unit)} \\ S_3 & n = 2 \text{ (Dyadic)} \\ S_4 & n = 3 \text{ (Triadic)} \\ S_1 & \Omega(n) = 1,\ n \equiv 1 \pmod 6 \text{ (Solar)} \\ S_2 & \Omega(n) = 1,\ n \equiv 5 \pmod 6 \text{ (Lunar)} \\ S_5 & \Omega(n) = 2 \text{ (Semiprime)} \\ S_6 & \Omega(n) \geq 3 \text{ (Multiprime, absorbing)} \end{cases}$$

Table 1: The seven classes with examples and density at $N = 10{,}000$.

Class	Name	Definition	Examples	N=10,000
S_0	Unit	$n = 1$	1	0.01%
S_1	Solar	prime, $p \equiv 1\,(6)$	7,13,19	6.1%
S_2	Lunar	prime, $p \equiv 5\,(6)$	5,11,17	6.2%
S_3	Dyadic	$n = 2$	2	0.01%
S_4	Triadic	$n = 3$	3	0.01%
S_5	Semiprime	$\Omega(n) = 2$	4,6,9	26.2%
S_6	Multiprime	$\Omega(n) \geq 3$	8,12,24	61.5%

Proposition 3.2 (Exhaustive partition)**.** cl *defines a partition of $\mathbb{N}$: every natural number belongs to exactly one class.*

Proof. The cases are hierarchically disjoint: $n = 1, 2, 3$ tested first; remaining primes classified by $\Omega = 1$ and residue mod 6 (Proposition 2.3 guarantees all primes > 3 land in S_1 or S_2); then $\Omega = 2 \to S_5$, $\Omega \geq 3 \to S_6$. □ □

Remark 3.3 (Density structure)**.** *All classes except S_6 have natural density 0. For $N \geq 100$, more than 60% of integers are Multiprimes.*

3.2 The Multiplicative Monoid

Definition 3.4. *For $C_1, C_2 \in S$, define $C_1 \otimes C_2 := \mathrm{cl}(n_1 \cdot n_2)$ for any $n_i \in C_i$.*

Theorem 3.5 (Well-definedness)**.** *The operation $\otimes$ is well-defined: $\mathrm{cl}(n_1 n_2)$ depends only on $\mathrm{cl}(n_1)$ and $\mathrm{cl}(n_2)$.*

Proof. The singleton classes $S_0 = \{1\}$, $S_3 = \{2\}$, $S_4 = \{3\}$ have unique representatives, so products involving them are trivially well-defined. For non-singleton classes (S_1, S_2, S_5, S_6), note that $\Omega(n_1 n_2) = \Omega(n_1) + \Omega(n_2) \geq 2$ by Proposition 2.2, so the product never falls into the $\Omega = 1$ range where the mod-6 residue distinction applies. The output class is therefore determined entirely by the Ω-sum: $\Omega = 2 \to S_5$; $\Omega \geq 3 \to S_6$. Since Ω is the sole determinant for classes with $\Omega \geq 2$, the map $\mathrm{cl}(n_1 n_2)$ depends only on $\mathrm{cl}(n_1)$ and $\mathrm{cl}(n_2)$. □ □

Theorem 3.6 (Complete Cayley table)**.** *The multiplication $\otimes$ on S is given by Table 2.*

Theorem 3.7 (Monoid structure [10])**.** *$(S, \otimes)$ is a finite commutative monoid:*

1. Associativity: $(C_1 \otimes C_2) \otimes C_3 = C_1 \otimes (C_2 \otimes C_3)$.
2. Neutral element: $S_0 \otimes C = C$ *for all* C.
3. Absorbing element: $S_6 \otimes C = S_6$ *for all* C.
4. Commutativity: $C_1 \otimes C_2 = C_2 \otimes C_1$.
5. Idempotents: $\{S_0, S_6\}$ *only.*

Proof. Properties (2)–(5) are verified by inspection of Table 2. For (1): pick representatives $a \in C_1$, $b \in C_2$, $c \in C_3$. Then $(C_1 \otimes C_2) \otimes C_3 = \mathrm{cl}(r \cdot c)$ for any $r \in \mathrm{cl}(a \cdot b)$. By Theorem 3.5, $\mathrm{cl}(r \cdot c)$ depends only on $\mathrm{cl}(r)$ and $\mathrm{cl}(c)$, so $\mathrm{cl}(r \cdot c) = \mathrm{cl}((a \cdot b) \cdot c) = \mathrm{cl}(a \cdot (b \cdot c))$, where the last equality uses associativity of integer multiplication. By the symmetric argument, $C_1 \otimes (C_2 \otimes C_3) = \mathrm{cl}(a \cdot (b \cdot c))$. □ □

Theorem 3.8 (Maximality of the 7-class partition)**.** *Among all set partitions of $\{S_0, \ldots, S_6\}$, the identity partition (7 blocks) is the unique* maximal *element (in the refinement order) satisfying both*

(i) $\otimes$-closure: *for every pair of blocks (B, B'), the set $\{\mathrm{cl}(ab) : a \in B, b \in B'\}$ lies in a single block (well-definedness of the induced multiplication, Theorem 3.5);*

(ii) Preservation of the 7 deterministic addition laws *of Table 4: each law $\{S_i, S_j\} \to S_k$ remains deterministic under the quotient, with output block $[S_k]$.*

Equivalently, no refinement of the 7-class partition exists within the invariants $(\Omega, p \bmod 6)$, and every strict coarsening either violates (i) or loses at least one deterministic addition law.

Proof. (i) is Theorem 3.5; for the identity partition, (ii) holds by inspection of Table 4. For the converse, we enumerate all Bell(7) = 877 partitions of $\{S_0, \ldots, S_6\}$ exhaustively (experiment E1.0b): exactly 7 partitions satisfy both (i) and (ii), and these form a single chain in the refinement order

$$\pi_1 \prec \pi_2 \prec \pi_3 \prec \pi_4 \prec \pi_5 \prec \pi_6 \prec \pi_7$$

with π_7 the identity. In particular π_7 is the unique maximal element; every strict coarsening (any of $\pi_1, \ldots, \pi_6$, or any of the remaining 870 partitions) either fails (i) or collapses at least one deterministic addition law of Table 4. Full data in `results/exp_01_0b_partition_lattice.json`. □ □

Remark (multiplication-only sub-quotient). In the multiplicative direction alone, the four prime classes S_1, S_2, S_3, S_4 share identical Cayley rows, so the $\otimes$-closed 4-element quotient $\{S_0, S_{\text{prime}}, S_5, S_6\}$ also exists. It appears in the enumeration above as a $\otimes$-closed partition that is *not* among the 7 chain elements: it collapses $S_3 + S_3 \to S_5$ and $S_1 + S_1 \not\to$ constant into the same block pair, violating (ii). The full 7-class refinement is therefore necessary precisely when addition laws are considered.

3.3 Power Dynamics

Theorem 3.9 (Exponentiation trajectories)**.** *For $n \in C$ and $k \geq 1$: $\Omega(n^k) = k \cdot \Omega(n)$.*

The resulting trajectories are summarised in Table 3.

Theorem 3.10 (Universality of Multiprime)**.** *For every $n > 1$, there exists $k \in \mathbb{N}$ such that $\mathrm{cl}(n^k) = S_6$.*

Proof. Set $k = \lceil 3/\Omega(n) \rceil$. Then $\Omega(n^k) = k \cdot \Omega(n) \geq 3$, giving $\mathrm{cl}(n^k) = S_6$. □ □

Verification. 99,999 integers $n \in [2, 100{,}000]$: all reach S_6; maximum observed $k = 3$, matching the theoretical maximum $\lceil 3/1 \rceil = 3$ for primes.

Interpretation. Exponentiation drives every number toward the Multiprime basin. The flow $S_0 \to$ Primes $\to S_5 \to S_6$ is irreversible under repeated self-multiplication.

Table 2: Cayley table of $(S, \otimes)$. Verified on 200 pairs per entry (9,800 total), zero deviations (E7.1).

$\otimes$	S_0	S_1	S_2	S_3	S_4	S_5	S_6
S_0	S_0	S_1	S_2	S_3	S_4	S_5	S_6
S_1	S_1	S_5	S_5	S_5	S_5	S_6	S_6
S_2	S_2	S_5	S_5	S_5	S_5	S_6	S_6
S_3	S_3	S_5	S_5	S_5	S_5	S_6	S_6
S_4	S_4	S_5	S_5	S_5	S_5	S_6	S_6
S_5	S_5	S_6	S_6	S_6	S_6	S_6	S_6
S_6	S_6	S_6	S_6	S_6	S_6	S_6	S_6

Table 3: Power dynamics $\mathrm{cl}(n^k)$ by starting class.

Class C	$k=1$	$k=2$	$k \geq 3$
S_0	S_0	S_0	S_0
Primes (S_1, S_2, S_3, S_4)	C	S_5	S_6
S_5	S_5	S_6	S_6
S_6	S_6	S_6	S_6

4 Addition Laws

4.1 Deterministic Laws

Definition 4.1. *An addition law $\{C_1, C_2\} \to C_3$ is* mathematically deterministic *if $\mathrm{cl}(n_1 + n_2) = C_3$ for* ***all*** *$n_1 \in C_1$, $n_2 \in C_2$ without exception.*

Definition 4.2 (Trivial vs. non-trivial determinism)**.** *A deterministic addition law $\{C_1, C_2\} \to C_3$ is* trivial *if at least one of C_1, C_2 is a singleton class (S_0=$\{1\}$, S_3=$\{2\}$, or S_4=$\{3\}$), in which case determinism reduces to an arithmetic identity between two fixed integers. Otherwise both operand classes are infinite and determinism is an emergent statistical property; such laws are called* non-trivial.

Theorem 4.3 (Deterministic addition laws)**.** *Among the $\binom{7}{2} + 7 = 28$ unordered class pairs, exactly* ***one non-trivial*** *and* ***six trivial*** *deterministic addition laws exist (Definition 4.2, Table 4). The unique non-trivial law is Solar+Lunar (Theorem 4.4).*

Count. There are $\binom{7}{2} + 7 = 28$ unordered class pairs. Non-determinism of each of the remaining 21 is established by explicit counterexample.[1] Table 5 gives the full empirical distributions.

Theorem 4.4 (Solar + Lunar $\to$ Multiprime)**.** *For any Solar prime $p \in S_1$ and Lunar prime $q \in S_2$: $\mathrm{cl}(p + q) = S_6$.*

[1]For instance, $\{S_0, S_2\}$ might appear deterministic empirically ($P(S_6) \approx 62.5\%$ in Table 5), but is refuted by a single witness: $1 + 5 = 6$ with $\mathrm{cl}(6) = S_5$, while $1 + 11 = 12$ with $\mathrm{cl}(12) = S_6$. Experiment E-S0.S2 in the repository confirms that every non-deterministic pair admits at least two distinct output classes.

Table 4: The 7 deterministic addition laws, split by Definition 4.2.

C_1	C_2	Result	Type	Justification
S_0	S_0	S_3	trivial	$1 + 1 = 2$
S_0	S_3	S_4	trivial	$1 + 2 = 3$
S_0	S_4	S_5	trivial	$1 + 3 = 4 = 2^2$
S_3	S_3	S_5	trivial	$2 + 2 = 4 = 2^2$
S_3	S_4	S_2	trivial	$2+3 = 5$ (Lunar)
S_4	S_4	S_5	trivial	$3 + 3 = 6 = 2\times3$
$\mathbf{S_1}$	$\mathbf{S_2}$	$\mathbf{S_6}$	**non-trivial**	**Thm. 4.4**

Proof. $p \equiv 1 \pmod 6$ and $q \equiv 5 \pmod 6$ imply $p+q \equiv 0 \pmod 6$, so $6 \mid (p + q)$. Since $p \geq 7$ and $q \geq 5$, we have $p + q \geq 12$. Write $p + q = 6m$ with $m \geq 2$. Then $\Omega(p+q) = \Omega(6m) = \Omega(6)+\Omega(m) = 2+\Omega(m) \geq 3$, giving $\mathrm{cl}(p + q) = S_6$. □ □

Verification (E4.1). All 90,000 Solar–Lunar pairs in $[7, 100{,}000]$: zero exceptions (95% CI for $P(S_6)$: $[0.99996, 1.0]$).

4.2 Probabilistic Laws

For the remaining 21 class pairs, $\mathrm{cl}(n_1 + n_2)$ varies with representative choice. Table 5 characterises all 21 distributions empirically.

The symmetric Solar/Lunar structure ($P(S_6 \mid S_1{+}S_1) \approx P(S_6 \mid S_2{+}S_2) \approx 88.4\%$) confirms the Dirichlet balance (Section 2.2). The anomalous distribution of $S_2 + S_3$ (only 33.8% S_6, 23.9% prime) arises from the special role of 2: adding 2 to a Lunar prime $q \equiv 5 \pmod 6$ gives $q + 2 \equiv 1 \pmod 6$, frequently yielding another prime.

Table 5: Empirical distributions for all 21 probabilistic addition laws ($n = 100{,}000$ pairs each, 95% CI width <0.007).

C_1	C_2	$P(S_6)$	$P(S_5)$	$P(\text{prime})$
S_0	S_1	62.2%	31.1%	6.7%
S_0	S_2	62.5%	31.0%	6.5%
S_0	S_5	74.4%	19.2%	6.4%
S_0	S_6	71.6%	20.8%	7.6%
S_1	S_1	88.4%	11.6%	<0.1%
S_1	S_3	82.5%	17.5%	0%
S_1	S_4	86.8%	13.2%	0%
S_1	S_5	88.3%	10.8%	0.9%
S_1	S_6	86.5%	12.3%	1.2%
S_2	S_2	88.4%	11.6%	<0.1%
S_2	S_3	33.8%	42.3%	23.9%
S_2	S_4	79.8%	18.9%	1.3%
S_2	S_5	87.4%	11.2%	1.4%
S_2	S_6	86.5%	12.2%	1.3%
S_3	S_5	74.1%	25.9%	0%
S_3	S_6	72.5%	21.0%	6.5%
S_4	S_5	79.2%	17.3%	3.5%
S_4	S_6	72.8%	20.7%	6.5%
S_5	S_5	87.5%	12.5%	0%
S_5	S_6	87.0%	11.5%	1.5%
S_6	S_6	71.8%	20.6%	7.6%

5 SpectralAddress: Numbers as Coordinates

5.1 Definition

Definition 5.1 (SpectralAddress)**.** *Fix a prime basis* $\mathbf{B} = (p_1, \ldots, p_K)$ *with* p_i *the* i*-th prime. The SpectralAddress of* n *is:*

$$\mathrm{SA}(n) = \big(\nu_{p_1}(n), \ldots, \nu_{p_K}(n)\big) \in \mathbb{N}_0^K$$

Default basis: $K = 7$, $\mathbf{B} = (2, 3, 5, 7, 11, 13, 17)$.

Limitation. SpectralAddress is lossless only for $\mathbf{B}$-smooth integers [13]. For $K = 7$ and n up to $17^7 = 410{,}338{,}673$, the representation is exact.

5.2 Algebraic Isomorphisms

Proposition 5.2 (Multiplication isomorphism)**.** *For* $\mathbf{B}$*-smooth* a, b*:*

$$\mathrm{SA}(ab) = \mathrm{SA}(a) + \mathrm{SA}(b) \quad \textit{(component-wise, } O(K)\textit{)}$$

Proposition 5.3 (GCD and LCM isomorphisms)**.** *For* $\mathbf{B}$*-smooth* a, b*:*

$$\mathrm{SA}(\gcd(a,b)) = \min(\mathrm{SA}(a), \mathrm{SA}(b)) \quad \textit{(}O(K)\textit{)}$$
$$\mathrm{SA}(\mathrm{lcm}(a,b)) = \max(\mathrm{SA}(a), \mathrm{SA}(b)) \quad \textit{(}O(K)\textit{)}$$

Proposition 5.4 (Coprimality criterion)**.** *For* $\mathbf{B}$*-smooth* a, b*:*

$$\gcd(a,b) = 1 \iff \mathrm{SA}(a) \cdot \mathrm{SA}(b) = 0$$

Warning (scope restriction). *This criterion holds* only *for* $\mathbf{B}$*-smooth integers. If* a *or* b *carries a prime factor outside* $\mathbf{B}$*, the SA-vectors cannot detect it, yielding a false positive.* Example: $a = 2 \cdot 19 = 38$, $b = 3 \cdot 19 = 57$ *with the default basis give inner product* 0*, yet* $\gcd(38, 57) = 19 \neq 1$.

Table 6: Arithmetic operations as geometry.

Arithmetic operation	SA operation	Complexity
Multiplication $a \times b$	Vector addition	$O(K)$
$\gcd(a,b)$	Component-wise min	$O(K)$
$\mathrm{lcm}(a,b)$	Component-wise max	$O(K)$
Coprimality test	Inner product $= 0$?	$O(K)$
Classification $\mathrm{cl}(n)$	L_1 norm of $\mathrm{SA}(n)$	$O(K)$

Verification. Bijection: 10,000/10,000 round-trip reconstructions exact. Four isomorphisms: 40,000/40,000 tests passed.

6 Resonance Relation and Sparsity

6.1 Definition

Definition 6.1 (Resonance)**.** *For integers i, j:*

$$R(i,j) = \|\min(\mathrm{SA}(i), \mathrm{SA}(j))\|_1 = \Omega(\gcd(i,j))$$

We call i and j resonant *if $R(i,j) > 0$, i.e., $\gcd(i,j) > 1$. The term* resonance *denotes structural co-divisibility; it has no topological or analytic content beyond this arithmetic definition.*

6.2 Asymptotic Sparsity

Theorem 6.2 (Coprime density)**.**

$$P\big(R(i,j) = 0\big) = P\big(\gcd(i,j) = 1\big) = \frac{6}{\pi^2} \approx 0.6079$$

Proof. Classical: $P(\gcd(i,j) = 1) = \prod_p (1 - p^{-2}) = 1/\zeta(2) = 6/\pi^2$. □ □

Verification. Empirical coprime fractions for $T \in \{64, \ldots, 8192\}$ converge monotonically to $6/\pi^2$ (deviation at $T = 8{,}192$: 0.0002).

6.3 Multi-Scale Resonance Hierarchy

Experiment E6.2 ($T = 2{,}100$; 2,203,950 total pairs):

Table 7: Multi-scale resonance hierarchy (E6.2).

Type	Definition	Empirical	Theoretical
Type-2	both div. 2	0.2499	$\approx 1/4$
Type-23	both div. 6	0.02771	$\approx 1/36$
Type-235	both div. 30	0.001096	$\approx 1/900$
Type-2357	both div. 210	2.04×10^{-5}	$\approx 1/44100$

This logarithmic hierarchy constitutes a natural multi-scale structure arising from arithmetic. Experimental validation as a routing prior for sequence processing is ongoing and will be presented in a companion paper.

7 Computational Verification

7.1 Python Reference Implementation

All theoretical results are verified in Python via a shared classification module (`spectral_matrix.py`); only E4.1 requires `sympy` for symbolic verification. All 18 pytest-runnable experiments are listed in the repository's `EXPERIMENT_INDEX.md`.

- E0.0 (SpectralMemory benchmark): 2-layer, 4-head architecture on TinyShakespeare [15]; 5 tests (context scaling, latency, memory, infinite retrieval, sliding-window comparison); see Section 8.3
- E1.0 (Maximality of 7-class partition, illustrative): confirmed via three-step argument (multiplicative equivalence of $S_1 \equiv S_2 \equiv S_3 \equiv S_4$, additive $S_1 \not\equiv S_2$, vacuity of S_5/S_6 splits)
- E1.0b (Maximality of 7-class partition, exhaustive): all Bell(7)=877 set partitions of $\{S_0, \ldots, S_6\}$ enumerated; 68 are $\otimes$-closed, 7 preserve all 7 (1 non-trivial + 6 trivial, Definition 4.2) deterministic addition laws, and these form a chain with the identity 7-partition as the unique maximum (Theorem 3.8)
- E1.1 (Ω-additivity): 100,000 random pairs, rate 1.0
- E1.2 (Prime residues mod 6): all 664,579 primes below 10^7
- E2.1 (Partition completeness): verified exhaustive coverage
- E2.3a (Coprime sparsity): empirical $6/\pi^2$ convergence
- E2.3b (Monoid identity/absorption): verified S_0, S_6 properties
- E3.1 (Power dynamics): 99,999 integers, max $k = 3$
- E4.1 (Solar+Lunar, `sympy`): 90,000 pairs, zero exceptions
- E5.1 (SA bijection): 10,000 round-trips, zero failures
- E5.2 (SA isomorphisms): 40,000 tests, 100% pass rate
- E5.3a (Classification recovery from SA): 100,000 randomly generated **B**-smooth values (basis $\{2, 3, 5, 7, 11, 13, 17\}$, exponents 0–4) yielded 56,282 distinct integers (birthday collisions due to finite sample space 5^7=78,125); cl(n) via direct factorization agrees with cl recovered from SA(n) in 100% of cases (0 disagreements on all 56,282 unique values)
- E5.3b (**B**-smooth boundary): bijection breaks at $n = 19$ (first prime $> \max(\mathbf{B})$); **B**-smooth fraction decays from 28.7% at N=1,000 to 9.45% at N=10,000 to 2.58% at N=100,000
- E6.2 (Resonance hierarchy): 2,203,950 pairs at $T = 2{,}100$
- E7.1 (Cayley table): 200 pairs/entry (9,800 total), zero deviations
- E10.0 (All 28 addition laws): 100,000 pairs each; 1 non-trivial and 6 trivial (singleton-induced) deterministic laws confirmed, 21 probabilistic with explicit witnesses

7.2 C++20 High-Performance Implementation

[a] Precomputed class indices; 559.9 M ops/s on AMD Ryzen 9 3900 (GCC 14.2, `-O3 -march=native`).

Table 8: C++20 benchmarks (Intel Core i5, `-O3`).

Operation	Throughput	Complexity
cl(n) classification	0.84 M ops/s	$O(\sqrt{n})$
Cayley lookup	475 M ops/s[a]	$O(1)$
SA computation	19.9 M ops/s[b]	$O(K)$
SA GCD	1,140 M ops/s[c]	$O(K)$
SA round-trip	10,000/10,000	—

[b] Includes trial division over $K = 7$ basis primes; reference value measured on Intel Core i5. Reproduction on AMD Ryzen 9 3900 (2 cores in container, GCC 14.2, `-O3 -march=native`) yields 16.0 M ops/s (within hardware variance, single-core throughput-bound).
[c] Component-wise minimum on *precomputed* SA vectors (memory-bandwidth-bound, K=7 component-wise min); measured 1,140.3 M ops/s on AMD Ryzen 9 3900 (GCC 14.2, `-O3 -march=native`). Combined SA-compute+GCD is bounded by the $O(K)$ trial-division in [b]; the GCD step itself is near memory bandwidth.

The Cayley table fits in 49 bytes; a single lookup requires one array access.

7.3 Verilog RTL Proof of Concept

A Verilog module `spectral_cell.v`, intended for synthesis, implements SpectralAddress lookup for $n < 1024$ via a precomputed BRAM18 lookup table ($1024 \times 21 = 21{,}504$ bits ≈ 2.6 KB):

```
module spectral_cell #(parameter K=7) (
    input  clk,
    input  [9:0]   n_in,
    output reg [K*3-1:0] sa_out
);
  reg [K*3-1:0] lut [0:1023];
  initial $readmemb("sa_lut.mem", lut);
  always @(posedge clk) sa_out <= lut[n_in];
endmodule
```

Proof-of-concept results: 14/14 test vectors passed (Python behavioral emulation of the RTL design). This constitutes a functional proof of concept, not exhaustive hardware verification. RTL synthesis, place-and-route, timing simulation, and full-coverage testing (all 1,024 inputs) remain pending. Target platform: Xilinx Artix-7 (Arty A7-35T); latency of 1 clock cycle at 100 MHz is the design target, not a verified timing result.

8 Computational Framework

8.1 Operational Summary

The individual components (Ω-function, mod-6 residues, p-adic valuations) are classical. The contribution of this framework is their combination into a single classification scheme with a complete algebraic characterization, deterministic addition laws, and systematic operationalization as a computational toolkit.

Concept	Classical view	PrimeSpec oper.
What is n?	A count	Coord. SA(n)
Connects i, j?	Diff. $\lvert i-j \rvert$	Resonance $R(i,j)$
What is prime?	Not divisible	Atom: 1 axis
Multiply?	Rpt. addition	Vec. add. $O(K)$
What is GCD?	Largest divisor	Min component

Table 9: Classical view vs. PrimeSpec operationalization.

8.2 Informational Content

The classification entropy at $N = 10{,}000$ (computed from the empirical densities in Table 1):

$$H(S) = -\sum_{i=0}^{6} P(S_i) \log_2 P(S_i) \approx 1.47 \text{ bits per integer}$$

This lies well below the maximum $\log_2 7 \approx 2.807$ bits, reflecting the strong dominance of S_6 (61.5% of all integers at this range).

The SpectralAddress at $K = 7$ carries ≈ 6.17 bits per vector (Shannon entropy of the coordinate distribution over the first 10^4 integers).

8.3 Structural Computation

For associative memory. A query at position q resonates with all positions p where $\gcd(q, p) > 1$. With a precomputed SpectralAddress table, computing resonance scores for N candidate positions requires $O(N \cdot K)$ arithmetic operations.

Experiment E0.0 (SpectralMemory benchmark). A 2-layer, 4-head SpectralMemory architecture replaces learned positional encodings with p-adic valuation vectors and replaces the causal attention mask with the resonance AND-gate $R(i,j) = \sum_p \min(\nu_p(i), \nu_p(j))$. No positional parameters are learned; all routing derives from prime structure. The architecture is benchmarked against standard causal attention and a sliding-window baseline on TinyShakespeare across five tests:

1. *Context scaling* ($T \in \{128, 256, 512\}$): The resonance-based mask achieves ≈61% sparsity (matching the $6/\pi^2$ coprime fraction). The validation loss gap closes monotonically with context length, reaching parity at $T = 512$ ($\Delta_{\text{val}} = -0.022$).
2. *Latency scaling*: Wall-clock time per forward pass as a function of T.

3. *Memory scaling*: Peak memory allocation as a function of T.
4. *Infinite retrieval*: Resonance-based retrieval of a target token at distances 10,000 to 500,000 positions runs in $O(K)$ time independent of distance.
5. *Sliding-window comparison*: Direct comparison with Longformer-style fixed-window attention.

Status and scope. These results demonstrate the operational feasibility of resonance-based attention routing but are limited to a small-scale setting (TinyShakespeare, 2-layer model, CPU). Large-scale validation on production datasets and GPU hardware is deferred to a companion paper. No performance superiority claims are made in the present work.

For database indexing. Records indexed by **B**-smooth key n have SpectralAddress $\mathrm{SA}(n)$. Divisibility queries reduce to single-axis lookups; structural similarity queries reduce to K-comparison resonance tests. The **B**-smoothness restriction must be verified for all keys.

For computational hardware. The entire Cayley table fits in 49 bytes; SpectralAddress lookup for $n < 1024$ fits in ≈ 2.6 KB BRAM. The resonance gate is a single bitwise AND. These operations are hardware-native at commodity FPGA resource costs.

9 Related Work and Open Problems

9.1 Positioning

The individual components (Ω-function, mod-6 residues, p-adic valuations, coprime density) are classical, and contributions C1, C2, C5 systematize known results into a unified framework. The novel content is the complete characterization of deterministic addition laws (C3), in particular the Solar+Lunar law (Theorem 4.4), and the combined operationalization as a computational toolkit with cross-platform verification. To our knowledge, no prior work has combined these components into a single classification scheme with (1) a complete Cayley table, (2) a full characterization of deterministic addition laws with explicit non-determinism witnesses, or (3) cross-platform computational verification including hardware behavioral simulation. Geroldinger and colleagues study non-unique factorization in monoids [3]. p-adic applications in ML [9] use ultrametric topology; our SpectralAddress provides a compatible finite-dimensional substrate. For computational aspects of primality and factorization see [12]; for additive number theory see [11].

9.2 Limitations

1. SpectralAddress is exact only for **B**-smooth integers; non-smooth factors require extended basis or explicit treatment.
2. The monoid's essential structure is 4-element; the 7-class refinement is necessary for addition laws and SpectralAddress but not for multiplication alone.
3. ML application results are preliminary; large-scale validation is ongoing and deferred to a companion paper. No performance claims are made in the present work.

9.3 Open Problems

1. **Complete addition characterization.** Find closed-form distributions for the 21 non-deterministic addition laws.
2. **Finer partitions.** Does there exist a partition based on Ω and residues mod $m > 6$ that is closed under multiplication and admits a strictly larger set of non-trivially deterministic addition laws?
3. **Gaussian integer extension.** Does an analogous semigroup exist over $\mathbb{Z}[i]$?
4. **Optimal basis size.** How does information content and computational utility scale with K?
5. **Prime pair sums in S_6.** Does every sufficiently large n with $6 \mid n$ arise as a sum $p + q$ with $p \equiv 1 \pmod 6$ and $q \equiv 5 \pmod 6$ both prime? This is *strictly stronger* than Goldbach's conjecture restricted to multiples of 6, since it additionally imposes residue conditions on the summands. Standard circle-method results [11] apply to the unconstrained case; the residue-constrained variant appears open.
6. **L-function connections.** Are there relationships between the monoid structure and the analytic theory of the Selberg class [14]?

10 Conclusion

We have completely characterised the algebraic structure induced by classifying $\mathbb{N}$ via Ω-depth and prime residue mod 6.

By making prime-arithmetic structure explicit, computable, and hardware-native, we enable systems that reason about numbers not only as ordered magnitudes but as elements of a structured algebraic space.

Code and data: `github.com/Prime-Spectrum-Team/Numbers-are-Information-`

Result	Statement	Status
7-class partition	Exhaustive partition	Proved
Monoid	Finite comm. monoid	Proved
Cayley table	49 entries verified	Proved+ver.
Universality	$n>1$ reaches S_6	Proved+ver.
7 add. laws	1 non-triv. + 6 triv. det.	Proved
Solar+Lunar	$S_1 + S_2 \to S_6$	Proved
SpectralAddress	4 isom., $O(K)$	Proved+ver.
Resonance	$6/\pi^2$ non-res.	Proved+ver.
Verilog RTL	PoC 14/14 vectors	RTL pending

Table 10: Summary of contributions and verification status.

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A Python Reference Classifier

```
def omega(n):
    count, d = 0, 2
    while d * d <= n:
        while n % d == 0:
            count += 1; n //= d
        d += 1
    return count + (1 if n > 1 else 0)

def classify(n):
    if n == 1: return 'S0'
    if n == 2: return 'S3'
    if n == 3: return 'S4'
    w = omega(n)
    if w == 1: return 'S1' if n % 6 == 1 else 'S2'
    if w == 2: return 'S5'
    return 'S6'

def spectral_address(n, basis=(2,3,5,7,11,13,17)):
    sa = []
    for p in basis:
        v = 0
        while n % p == 0: v += 1; n //= p
        sa.append(v)
    return tuple(sa)

CAYLEY = [
    [0,1,2,3,4,5,6],
    [1,5,5,5,5,6,6],
    [2,5,5,5,5,6,6],
    [3,5,5,5,5,6,6],
    [4,5,5,5,5,6,6],
    [5,6,6,6,6,6,6],
    [6,6,6,6,6,6,6],
]
```